PROGRESSIONS 6

» Evidence is found that Babylonians some 400years ago, knew of arithmetic and geometric progressions.

» Among the Indian mathematicians, Aryabhatta (470 AD) was the first to give formula for the sum of squares and cubes of natural numbers in his famous work "Aryabhatiyam"

» Indian mathematician Brahmagupta (598 AD), Mahavira (850 AD) and Bhaskara (1114-1185 AD) also considered the sums of squares and cubes.

□ ***Arithmetic Progression (A.P.)***

An arithmetic progression (A.P.) is a list of numbers in which each term is obtained by adding a fixed number "d" to the preceding term, except the first term. The fixed number "d" is called the 'common difference'.

Ex: 1, 4, 7, 10 13…. are in A.P.; Here d = 3

» Let $a_1, a_2, a_3, \ldots a_k, a_{k+1}, \ldots a_n \ldots$ be an A.P.
Let its common difference be d, then
$d = a_2 - a_1 = a_3 - a_2 = \ldots = a_{k+1} - a_k = \ldots$

» If the first term is 'a' and the common difference is "d" then a, a + d, a + 2d, a + 3d, … is an A.P.

□ ***General term of an A.P.***

» Let "a" be the first term and "d" be the common difference of an A.P. Then, its nth term or general term is given by $a_n = a + (n - 1) d$

Ex: The 10th term of the A.P. given by 5, 1, −3, −7, …. is $a_{10} = 5 + (10 - 1)(-4) = -31$

» If the number of term of an A.P. is finite, then it is a finite A.P. ***Ex:*** 13, 11, 9, 7, 5

» If the number of terms of an A.P. is infinite, then it is an infinite A.P. ***Ex:*** 4, 7, 10, 13, 16, 19….

» Three numbers in A.P. should be taken as

» a −d, a, a + d.
Four numbers in A.P. should be taken as

» a −3d, a − d, a + d, a + 3d.
Five numbers in A.P. should be taken as
a − 2d, a − d, a, a + d, a + 2d

» Six numbers in A.P. should be taken as
a − 5d, a − 3d, a − d, a + d, a + 3d, a + 5d

» If a, b, c a re in A.P.,then $b = \frac{a+c}{2}$ is called the arithmetic mean of "a" and "c".

» The sum of the first "n" terms of an A.P. is given by $s_n = n[2a + (n-1)d]$

» If the first and last terms of an A.P. are "a" and "*l*", the common difference is not given then $s_n = \frac{n}{2}(a + l)$

» $a_n = s_n - s_{n-1}$

» The sum of first "n" positive integers $s_n = \frac{n(n+1)}{2}$

Ex: sum of first "10" positive integers $= \frac{10(10+1)}{2} = 55$

□ ***Geometric Progression (G.P.)***

Ageometric progression is a list of numbers in which each term is obtained by multiplying preceding term with a fixed number "r" except first term. This fixed number is called common ratio "r"

Ex: 3, 9, 27, 81, … are in G.P.
Here common ratio r =3

» Alist of numbers $a_1, a_2, a_3, \ldots a_n \ldots$ are in G.P. then the commonratio
$r = \frac{a_2}{a_1} = \frac{a^3}{a_2} = \ldots\ldots = \frac{a_n}{a_{n-1}} = \ldots\ldots$

» The first term of a G.P. by "a" and common ratio "r" then the G.P. is a, ar, $ar^2, \ldots\ldots$

» If the first term and common ratio of a G.P. is a, r respectively then n^{th} term $a = ar^{n-1}$

2 Mark Questions

1. Write an A.P. having 4 as the first term, −3 as the common difference.

Sol: The A.P. with first term a and common difference d is given by a, a + d, a + 2d, a + 3d,….

Here a = 4, d = −3

So, the A.P. is

4, 4+(−3), 4 + 2(−3), 4 + 3(−3),

….. 4, 1, −2, −5, ….

2. Which term of the sequence −1, 3, 7, 11, ….is 95?

Sol: a = −1, d = 3−(−1) =4

Let 95 be the n^{th}term of the given A.P. then a_n=95

a + (n −1)d =95

−1+ (n − 1) 4 = 95 ⇒4(n−1) = 96

⇒n−1=96=24

4

n =25

Thus, 95 is 25^{th}term of the given sequence

3. Write the G.P. if the first term a = 3, and the common ratio r = 2.

Sol:Since 'a' is the first term it can easily be written

We know that in G.P. every succeeding term is obtained by multiplying the preceding term with common ratio 'r'. So to get the second term we have to multiply the first term a = 3 by the com- mon ratio r = 2.

∴ Second term = ar = 3 × 2 = 6

(□First term x common ratio)

Similarly the third term

= second term× common ratio = 6 × 2 = 12

If we proceed in this way we get the following G.P.

3, 6, 12, 24,....

4. The 17^{th}term of an A.P. exceeds its 10^{th}term by 7. Find the common difference.

Sol:According to the question

$a^{17= a_{10}+ 7}$

a + 16d =a + 9d +7

16d −9d = 7 ⇒d =1

Hence, the common difference is 1

5. $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$ Given terms form an. A.P. or not. If they form an AP. Find the common difference and write three more terms

Sol:Given terms : $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$

$\sqrt{2}, \sqrt{2\times4}, \sqrt{2\times9}, \sqrt{2\times16}\ldots$

$\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \ldots$

$d = a_2 - a_1 = \sqrt{2} - 2\sqrt{2} = \sqrt{2}$

$a - a = 3\sqrt{2} - 2\sqrt{2} = \sqrt{2}$

$3 - 2 = 4\sqrt{2} - 3\sqrt{2} = \sqrt{2}$

Given terms form an A.P

Comon difference d $= \sqrt{2}$

$a = a + d = 4\sqrt{2} + \sqrt{2} = 5\sqrt{2}$

$a = a + d = 5\sqrt{2} + \sqrt{2} = 6\sqrt{2}$

$a = a + d = 6\sqrt{2} + \sqrt{2} = 7\sqrt{2}$

∴***Threemoreterms are*** $5\sqrt{2}, 6\sqrt{2}, 7\sqrt{2}$

6. If an A.P a_n= 6n+2. Find the common difference.

Sol: a_n= 6n+ 2

a1 = 6(1) + 2 = 8

a2= 6(2) + 2 = 14

Common difference d = a_2– a_1= 14 – 8 = 6

7. Find the sum of the first 12 terms of the AP–37, –33, –29, ...

Sol:Here a = –37, d = –33–(–37)= 4, n = 12

$s_n = \frac{n}{2}[2a+(n-1)d]$

$s^{12} = \frac{12}{2}[2(-37) + (12-1)4]$

=6 (−74 + 44) = 6 (−30) = −180

8. Given a = 7, a_{13}= 35, find d and s_{13}

Sol:a = 7, a_{13}= a + 12d = 35

$7+12d=35, d=\frac{35-7}{12}=\frac{28}{12}=\frac{7}{3}$

$s_n = \frac{n}{}[2a+(n-1)d]$

$s_{13} = \frac{13}{2}[2(7) + (13-1)\frac{7}{3}]$

$= \frac{13}{2}[14 + 28] = \frac{13}{2} \times 42 = 13 \times 21 = 273$

9. Which term of the G.P $\sqrt{3}, 3, 3\sqrt{3}$ ....***is***729?

Sol: Given G.P. $\sqrt{3}, 3, 3\sqrt{3}....$

$$\textbf{\textit{Here}} = a = \sqrt{3}, r = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\textbf{\textit{Let}}\ a_n = 729 \quad ar^{n-1} = 729$$

$$\sqrt{3}.\sqrt{3}^{n-1} = 729 = 3^6$$

$$\sqrt{3}^{1+n-1} = {}^{6}\sqrt{3^n} = 3$$

$$(3^{1/2})^n = 3^6$$

$$\frac{n}{2} = 6, n = 12^3,$$

$\therefore$ 729 is the 12th term in G.P

4 Marks Questions

1. Subbarao started work in the year 2005 at an annual salary of Rs.5000, received Rs.200 raise each year. In which year did his annual salary will reach Rs. 7000?

Sol: Annual salary received by Subbarao in 2005, 2006, 2007 is Rs.5,000, Rs. 5,200, Rs. 5,400..... Clearly, it is in A.P.

Here a = 5,000, d = 200

Suppose Subbrao's annual salary reaches to Rs. 7,000 in n^{th} years. Then a_n = 7,000

a + (n−1)d = 7,000

5,000 + (n −1) 200 = 7,000

By simplifying n =11

Thus, 11th annual salary received by Subbarao will be Rs. 7,000. This means that after 10 years i.e. in the year 2015 his annual salary will reach to Rs. 7,000

2. Determine the A.P. Whose 3rd term is 5 and the 7th term is 9.

Sol: Given that 3rd term = a_3 = a + 2d = 5 → (1)

7th term = a_7 = a + 6d = 9 → (2)

subtraction

4d = 4

$$d = \frac{4}{4} = 1$$

Substitute d = 1 in eq (1), we get

a + 2 (1) = 5, a = 5–2, a = 3

Hence, the required A.P. is 3, 4, 5, ...

3. Which term of the AP: 3, 8, 13, 18, ...78?

Sol: Given terms in AP: 3, 8, 13, 18, ...

a = 3, d = 8 –3 = 5

Let a_n = 78

a +(n–1)d= 78

3 + (n–1) 5 = 78

$$n - 1 = \frac{78 - 3}{5} = 15$$

n = 15+1 = 16

$\therefore$ 16th term of A.Pis 78.

4. Find the 34 + 32 + 30 + ... +10

Sol: Given terms are in A.P.

Here a = 34, d = 32 –34 = –2

Let a_n = 10

a + (n – 1)d = 10

34 + (n – 1) (–2) = 10

$$n - 1 = \frac{10 - 34}{-2} = 12$$

n=13, $s_n = \frac{n}{2}[2a + (n-1)d]$

$\therefore$ sum of 13 terms

$$s_{13} = \frac{13}{2}[2(34) + (13-1)(-2)]$$

$$= \frac{13}{2}[68 - 24] = \frac{13}{2} \times 44 = 13 \times 22 = 286$$

$\therefore$ 34 + 32 + 30 + ... +10 = 286.

5. InG.P. the 3rd term is 24 and 6th term is 192. Find the 10 term.

Sol: In G.P. 3rd term = $ar^2 = 24 \rightarrow (1)$

6th term = $ar^5 = 192 \rightarrow (2)$

Dividing eq (2) eq (1)

$$\frac{ar^5}{ar^2} = \frac{192}{24} = 8$$

$$r^3 = 2^3, r = 2$$

Substitute r = 2 in eq (1), we get

$$a(2)^2 = 24, a = \frac{24}{4} = 6$$

$$10th\ term = ar^9 = (6)(2)^9 = 3072$$

6. Check whether – 150 is a term of the A.P. 11, 8, 5, 2, ...

Here a = 11, d = 8 –11 = – 3

Let a_n= – 150

a + (n–1) d = – 150

11 + (n – 1) (– 3) = – 150

$n-1=\frac{-150-11}{-3}=\frac{-161}{3}$

$n=\frac{-161}{3}+1=\frac{-161+3}{3}=\frac{-158}{3}$

But n should be a possible

So, – 150 is not a term of the given list numbers.

7. How many multiples of 4 lie between 10 and 250?

Sol: Multiples of 4 lie between 10 and 250 are

4, 8, 12, 16, ... 248

Above terms are in A.P.

Here a= 4, d = 8 – 4 = 4

Let a_n= 248

a + (n– 1) d = 248

4 + (n –1)4 = 248

$n-1=\frac{248-4}{4}=61$

n = 61+ 1 = 62

∴62nd term of A.P. is 248

There are 62, multiples of 4 lie between 10 and 250

8. Find the 20th term from the end of the A.P. 3, 8, 13, ... 253

Sol: A.P: 3, 8, 13, ...253

Here a = 3, d =8 –3 =5,

last term l= a_n= 253

an= 253, a + (n – 1)d = 253

3 + (n –1)5 = 253

$n-1=\frac{253-3}{5}=50$

n = 50+1 = 51.

So, there are 51 terms in the given A.P.

The 20th term from the end will be the 32nd term from the first term (∴ 51– 20 + 1 = 32)

∴ a_{32} = a + 31d = 3+31×5 = 158.

9. Find the sum of the first 40 positive integers divisible by 6

Sol: First 40 positive integers divisible by 6 are

6, 12, 18, ... 240

Here a = 6, d=12–6=6, n=40

$s_n=\frac{n}{2}[2a+(n-1)d]$

$s_{40}=\frac{40}{2}[2(6)+(40-1)6]$

= 20(12 + 234) = 4920

∴S40 = 4920.

6 Marks Questions

1. The sum of the 4th, 8thterms of an A.P. is 24, the sum of the 6th, 10h terms is 44. Find the first three terms of the A.P.

Sol: Given an A.P. in which

a4+ a_8= 24

(a + 3d) + (a + 7d) = 24

2a + 10d = 24

a + 5d = 12 →(1)

a6+ a_{10}= 44

(a + 5d) + (a + 9d) = 44

2a + 14d = 44

a + 7d = 22 →(2)

eq (2) − eq(1),

We get 2d =10 ⇒d = 5

From eq (1) a + 5×5 = 12

⇒a = −13

The first three terms of the

A.P. are a, a+d, a + 2d

⇒−13, −13 +5, −13 + 2(5)

⇒−3, −8, −3.

2. If the Geometric Progressions162, 54, 18…; $\frac{2}{81},\frac{2}{279},\frac{2}{...}$ havetheirn th term equal, find the value of n.

Sol: Given G.P. 162, 54, 18, …..

Here a =162

$r = \frac{a_2}{a_1} = \frac{54}{162} = \frac{1}{3}$

$a_n = ar^{n-1} = 162\left(\frac{1}{3}\right)^{n-1}$

$\frac{2}{81}, \frac{2}{27}, \frac{2}{8}, \ldots$

Here $a = \frac{2}{81}$

$r = \frac{a_2}{a_1} = \frac{2/27}{2/81} = 3$

$a_n = ar^{n-1} = \frac{2}{81}(3)^{n-1}$

Given that n^{th} terms are equal

$162\left(\frac{1}{3}\right)^{n-1} = \frac{2}{81}(3)^{n-1}$

$162 \times \frac{81}{2} = 3^{1-n} \times 3^{n-1}$

$81 \times 81 = 3^{2n-2}$

$34 \times 3^4 = 3^{2n-2}$

$\Rightarrow 3^{2n-2} = 3^{4+4} = 3^8$

$2n-2 = 8$

$n = 5$

$\therefore$ The 5th terms of the two G.P.s are equal.

3. If the 3rd and 9th terms of an A.P. are 4 and – 8 respectively. which term of this A.Pi Zero.

Sol: 3rd term of A.P= $a_3 = a + 2d = 4 \rightarrow (1)$

9th term of A.P= $a_9 = a + 8d = -8 \rightarrow (2)$

Subtracting

$a + 8d = -8$

$a + 2d = 4$

$- \quad - \quad -$

$6d = -12, d = -2$

Subtract $d = -2$ in eq (1), we get

$a + 2(-2) = 4, a = 4 + 4 = 8$

Let $a_n = 0$

$a + (n-1) = 0, 8 + (n-1)(-2) = 0$

$n - 1 = \frac{-8}{-2} = 4$

$n = 5$

5th term is zero

4. The fist and the last term of an A.P. are 17 and 350 respectively. If the common difference is 9. how many terms are there and what is their sum?

Sol: First term = 17

last term $l = 350$, common difference $d = 9$

Let $l = a_n = 350$

$a + (n-1)d = 350$

$17 + (n-1)9 = 350$

$n - 1 = \frac{350 - 17}{9} = 37$

$n = 38$

There are 38 terms in given A.P.

$S_n = \frac{n}{2}[a + l]$

$s_{38} = \frac{38}{2}[17 + 350] = 6973$